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Connection structure of oil hose for transmission

Abstract

A connection structure of a hose for transmission oil including a multilayer hose having a first rubber layer formed by a first pipe, a braided layer formed on an outer surface of the first rubber layer and formed by braiding aramid fibers. A second rubber layer located on the outer surface of the first rubber layer. A first connector including a body inserted into an inside of the first rubber layer and including a second pipe connected with the first pipe, and a stopper protruding from the body, and an inclined portion formed at an end of the body and forming an uphill slope of approximately 11°-17° to one side of the body. A second connector including a cover portion covering the multilayer hose and a plurality of pressurized portions formed concave with respect to an outer peripheral surface of the cover portion and in contact with the multilayer hose.

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Background/Summary

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Priority No. 10-2023-0165127 filed on Nov. 24, 2023, in the Korean Intellectual Property Office, the entire disclosure of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Technical Field

(2) Various embodiments disclosed in this document relate to the connection structure of an oil hose and transmission.

Description of Related Art

(3) In a power transmission element such as the transmission of vehicles, oil is used for the purpose of smooth operation, lubrication of mechanical friction elements, and cooling. In the case that the temperature of the oil used in the transmission is high, the oil may become acidic, which can reduce its durability.

(4) To prevent such occurrences, the vehicle is equipped with an oil cooling device for efficient cooling of the transmission oil.

(5) The above information may be provided as related art for the purpose of facilitating an understanding of this disclosure. No claim or decision is made as to whether any of the above can be applied as prior art in relation to this disclosure.

SUMMARY OF THE INVENTION

(6) In the case that the temperature of the oil used in the transmission is high, the oil may become acidic, which can reduce its durability. Inside the vehicle, an oil cooling device can be placed to cool the oil used in the transmission. Oil cooling device of transmission in the vehicle can form a fluid line through a hose. For example, a fluid line connecting an oil cooling device to a transmission can form a closed circulation loop. The overheated oil released from the outlet of the transmission can pass through the oil cooling device and enter the inlet of the transmission. In addition, inside the hose that connects the oil cooling device to the inlet of the transmission, the cooled oil through the oil cooling device can flow. The hose that connects the oil cooling device and the transmission can allow hot or cold oil to flow inside, depending on the installation location. Therefore, the hose needs to be formed of a material that is resistant to high and low temperatures.

In addition, since high pressure fluid can flow inside the hose, it is necessary to form a material that is resistant to high pressure.
(7) On the other hand, the hose may be exposed to vibration depending on the environment in which it is used. For example, a hose can vibrate while the vehicle is running. In general, a separate connector is used to connect a hose to another device or to another hose. Therefore, to maintain the connection with other hoses while the hose is exposed to vibration, it is preferred for the hose and the connector, and the other hose and the connector respectively to be tightly joined to each other.
(8) Technical problems to be solved in this document are not limited to the aforementioned technical problems, and other technical problems not described above may be easily understood from the following description by a person having ordinary knowledge in the art included in this document.
(9) The connection structure of a hose for transmission oil according to various embodiments disclosed in this document may include a multilayer hose including the first rubber layer formed by the first pipe, the braided layer formed on the outer surface of the first rubber layer and formed by braiding aramid fibers, and the second rubber layer located on the outer surface of the first rubber layer; a first connector including a body inserted into the inside of the first rubber layer and including a second pipe connected with the first pipe, a stopper protruding from the body, and an inclined portion formed at the end of the body and forming an uphill slope of approximately 11°-17° to one side of the body toward the stopper; and a second connector including a cover portion covering the multilayer hose and a plurality of pressurized portions formed concave with respect to the outer peripheral surface of the cover portion and in contact with the multilayer hose.
(10) According to various embodiments disclosed in this document, it is possible to disclose a multilayer hose that can be used under high pressure, extremely high temperature, and cryogenic condition and connect the transmission with the oil cooling device.
(11) According to various embodiments disclosed in this document, the connector connecting a multilayer hose to another multilayer hose may present a structure that is joined tightly to the multilayer hose. Therefore, the multilayer hose and the connector may maintain their coupling under external vibration. In addition, the tightly-joined structure of the multilayer hose and the connector may prevent the oil flowing inside the multilayer hose from spilling out to the outside of the multilayer hose.
(12) The effects obtainable in this disclosure are not limited to the aforementioned effects, and other effects not mentioned may be easily understood from the following description by a person having ordinary knowledge in the art to which the disclosure pertains.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) With regard to the description of the drawing, the same or similar reference numerals may be used for the same or similar components.
(2) FIG. **1**A, according to one embodiment of this disclosure, is a perspective view of a fluid line connecting an oil cooling device and a transmission.
(3) FIG. **1**B is an enlarged view of a portion of the fluid line of FIG. **1**A.
(4) FIG. **2** is a diagram of a multilayer hose connecting an oil cooling device and a transmission according to one embodiment of this disclosure.
(5) FIG. **3**A is a diagram of a connector assembly connecting different multilayer hoses according to one embodiment of this disclosure.
(6) FIG. **3**B shows an outer diameter of an end of a connector of the connector assembly of FIG. **3**A.
(7) FIG. **4**A is a diagram illustrating a portion of an assembly process for the connector assembly of

FIG. **3** according to one embodiment of this disclosure.
(8) FIGS. **4**B to **4**C are diagrams illustrating an assembly process of the connector assembly of FIG. **4**A during and after undergoing a swaging process according to one embodiment of this disclosure.
(9) FIG. **5**A is a diagram illustrating irregularities and an inclined portion formed at the end of the first connector according to one embodiment of this disclosure.
(10) FIG. **5**B is a cross-sectional view of a portion of the first connector of FIG. **5**A.
(11) FIG. **5**C is an enlarged view of an inclined portion of the first connector of FIG. **5**A.
(12) FIG. **6**A is a diagram illustrating a serration formed on the second connector according to one embodiment of this disclosure.
(13) FIG. **6**B is an enlarged view of a portion of the serration of FIG. **6**A.
DETAILED DESCRIPTION
(14) In the following description, various embodiments of this document are described with reference to the attached drawings. The various embodiments of this document and the terms used herein are not intended to limit the technological features set forth in this document to particular embodiments, but should be understood to include various changes, equivalents, or replacements for a corresponding embodiment.
(15) With regard to the description of the drawing, similar reference numerals may be used for similar or related elements. The singular form of a noun corresponding to an item may include one or more of the items, unless the relevant context clearly indicates otherwise.
(16) In this document, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. Such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). If an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively,” as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.
(17) FIGS. **1**A and **1**B, according to one embodiment of this disclosure, are perspective views of a fluid line connecting an oil cooling device and a transmission.
(18) According to one embodiment, the vehicle may include a power transmission element such as a transmission (e.g., a gearbox). A transmission may be a configuration that transmits power from the vehicle's engine to the wheels. In one embodiment, the vehicle may change gears and change the torque applied to the wheels via the transmission.
(19) According to one embodiment, oil may be used for the purpose of smooth operation, lubrication of mechanical friction elements, and cooling in a power transmission element such as a transmission. In the case that the oil used in the transmission is overheated, it may become acidic or lose its durability, which can cause the transmission to fail. To prevent such occurrences, the vehicle may be equipped with an oil cooling device for efficient cooling of the transmission.
(20) In one embodiment, with reference to FIGS. **1**A and **1**B, a fluid line connecting the oil cooling device and the transmission may be disposed inside the vehicle. The fluid line may connect the oil cooling device and the transmission and may consist of oil-flowing hoses **100** and **101**. In one embodiment, hoses **100** and **101** may each be multilayer hoses (e.g., a multilayer hose **100** of FIG. **2**) that may be used in high pressure, high temperature, and low temperature conditions. The fluid line may be a closed circulation loop in which the oil that is overheated in the transmission is cooled through an oil cooling device and then flows into the transmission.
(21) In one embodiment, with reference to FIGS. **1**A and **1**B, a plurality of hoses **100** and **101** including a fluid line may be connected via the connector assembly **200**. For example, one multilayer hose **100** (e.g., a multilayer hose **100** of FIG. **2**) and another multilayer hose **101** (e.g., a

multilayer hose **100** of FIG. **2**) may be connected via the connector assembly **200**. According to one embodiment, the connector assembly **200** may connect and seal plural multilayer hoses **100** and **101**. Therefore, the leakage of oil flowing inside the hoses **100** and **101** between the connector assembly **200** and the hoses **100** and **101** can be prevented. The description of the connector assembly **200** is described in detail in the drawings shown in FIGS. **3**A and **3**B.
(22) FIG. **2** is a diagram of a multilayer hose connecting an oil cooling device and a transmission according to one embodiment of this disclosure.
(23) The hoses **100** and **101** including the fluid line described above may each be of the same configuration as the multilayer hose **100** shown in FIG. **2**. Therefore, the description of the hoses **100** and **101** including the fluid line is to be replaced by the description of the multilayer hose **100** shown in FIG. **2**.
(24) According to one embodiment, as shown in FIG. **2**, the hose through which the transmission oil flows may consist of a multilayer hose **100**. In one embodiment, the multilayer hose **100** may include a first rubber layer **110** located on the inner side, a braided layer **120** formed by braiding on the outer surface of the first rubber layer **110**, and a second rubber layer **130** located on the outside of the multilayer hose **100**. The hose **101** may have a similar multilayer structure as described previously with respect to the multilayer hose **101** of FIG. **2**.
(25) In one embodiment shown in FIG. **2**, the first rubber layer **110** may include the first pipe **1001** through which the transmission oil flows. In one embodiment, the first connector **210** may be inserted into the inside of the first rubber layer **110**. In one embodiment, as described later, the first connector **210** may be inserted into the first pipe **1001** of the first rubber layer **110**. In this case, the inclined portion **214** of the first connector **210** may be the first part to be inserted into the first rubber layer **110** in the configuration of the first connector **210**. In one embodiment, the inner diameter D′**1** of the first rubber layer **110** may be formed smaller than the outer diameter D**1** of the outermost end of the body **211** of the first connector **210**. Therefore, the first rubber layer **110** and the first connector **210** may be joined tightly.
(26) In one embodiment, with reference to FIG. **2**, the braided layer **120** may be located on the outer surface of the first rubber layer **110**. In one embodiment, the braided layer **120** may be located between the outer surface of the first rubber layer **110** and the inner surface of the second rubber layer **130**.
(27) In one embodiment, the braided layer **120** may be formed along the outer peripheral surface of the first rubber layer **110** through a braiding process. The braiding process may be formed by inserting a mandrel (not shown) into the inside of the first rubber layer **110**, and then braiding the braided layer **120** using a braiding machine (not shown).
(28) In one embodiment, the braided layer **120** may consist of a plurality of braids. Braids may be made up of aramid fibers. An aramid fiber is a kind of synthetic fiber, which may have high strength and heat resistance.
(29) In one embodiment, the braided layer **120** may be a set of braids formed on the outer peripheral surface of the first rubber layer **110** through a braider. In one embodiment, the braided layer **120** may be a structure in which a plurality of braids are formed in a net shape. In one embodiment, one braid may be formed on the outer peripheral surface of the first rubber layer **110** with a pitch of about 30˜50 mm. The pitch can be the distance traveled when the braid is rotated for one turn. For example, the pitch may be the gap between the first point and the second point which is the same position as the first point when the braid is rotated at the first point of the first rubber layer **110**.
(30) In one embodiment, with reference to FIG. **2**, the second rubber layer **130** may be located on the outer surface of the first rubber layer **110**. In one embodiment, the second rubber layer **130** may be a configuration that forms the outer surface of the multilayer hose **100**.
(31) In one embodiment, the first rubber layer **110** and the second rubber layer **130** may be formed of a material that can be used under low temperature, high temperature and high pressure

conditions. The high pressure condition may be the state of an increased internal pressure of the multilayer hose **100**. In one embodiment, the first rubber layer **110** and the second rubber layer **130** may be formed from any of the following materials: acrylic rubber (ACM, Acrylate-chloroethyl vinylether monomer), ethylene acrylic rubber (AEM, ethylene acrylic elastomer), fluorinated rubber (FKM, fluorinated vinylidene monomer), and hydrogen nitrile rubber (H-NBR, hydrogenated acrylonitrile-butadiene rubber). In one embodiment, the first rubber layer **110** and the second rubber layer **130** may be formed through a plurality of combinations of acrylic rubber, ethylene acrylic rubber, fluorinated rubber, and hydrogen nitrile rubber.

(32) In one embodiment, acrylic rubber (ACM) may have a certain level of heat resistance, oil resistance, weather resistance, and ozone resistance. In one embodiment, acrylic rubber may be used in an environment of around −20° C. to 170° C.

(33) In one embodiment, ethylene acrylic rubber (AEM) may have a certain level of heat resistance, oil resistance, weather resistance, and ozone resistance. In one embodiment, ethylene acrylic rubber may be used in an environment of approximately −35° C. to 170° C.

(34) In one embodiment, hydrogen nitrile rubber (H-NBR) may have a certain level of abrasion resistance, low temperature resistance, oil resistance, fuel oil resistance, chemical resistance, permanent compression set resistance, heat resistance, and ozone resistance. In one embodiment, hydrogenated nitrile rubber may be used in an environment of about −40° C. to 140° C.

(35) In one embodiment, fluorinated rubber (FKM) may have a certain level of weather resistance, heat resistance, oil resistance, fuel oil, and chemical resistance. In one embodiment, fluorinated rubber may be used in an environment of about −20° C. to 250° C.

(36) According to one embodiment of this disclosure, the multilayer hose **100** may include a first rubber layer **110**, a braided layer **120**, and a second rubber layer **130**. In one embodiment, the multilayer hose **100** may be used in a high to low temperature state of approximately −40° C. to 150° C. and in a high pressure state of approximately 140 kpa to 2069 kpa as the first rubber layer **110** and the second rubber layer **130** are formed of any of the acrylic rubber, ethylene acrylic rubber, fluorinated rubber, and hydrogen nitrile rubber, and the braided layer **120** of aramid material is formed on the outer surface of the first rubber layer **110**.

(37) FIGS. **3**A and **3**B are diagrams of a first embodiment of a connector assembly connecting different multilayer hoses according to one embodiment of this disclosure. FIG. **4**A is a diagram illustrating an assembly process of the connector assembly of FIGS. **3**A and **3**B according to one embodiment of this disclosure. FIGS. **4**B to **4**C are diagrams illustrating an assembly process of the connector assembly of FIG. **4**A during and after undergoing a swaging process according to one embodiment of this disclosure. FIGS. **5**A to **5**C are diagrams illustrating irregularities and an inclined portion formed at the end of the first connector according to one embodiment of this disclosure. FIGS. **6**A and **6**B are diagrams illustrating a serration formed on the second connector according to one embodiment of this disclosure.

(38) According to one embodiment, as shown in FIGS. **3**A and **3**B, the multilayer hoses **100** and **101** connecting the transmission and the oil cooler may be connected via the connector assembly **200**. In one embodiment, the connector assembly may include a first connector **210** and a second connector **220**. In one embodiment, the first connector **210** may be configured to connect a plurality of multilayer hoses **100** and **101**. For example, one end **2101** (e.g., an end) of the first connector **210** may be inserted into one of the multilayer hoses **100**, and the other end **2102** may be inserted into another multilayer hose **101**. The second connector **220** may be a configuration that pressurizes the multilayer hose **100** inserted into the first connector **210**. That is, the second connector **220** may be configured to fix the multilayer hose **100** to the first connector **210**. The multilayer hose **100** may be joined tightly against the first connector **210** through the second connector **220**. Therefore, the multilayer hose **100** may not be separated from the first connector **210** in an environment where vibrations are applied. In addition, the leakage of oil flowing through the multilayer hose **100** into the gap between the multilayer hose **100** and the first connector **210**

may be prevented.
(39) In one embodiment, with reference to FIGS. **3**A and **3**B, the outer diameter D**1** of one end **2101** of the first connector **210** (e.g., the end **2101** of the first connector **210** or the end **2101** of the body **211** of the first connector **210**) may be larger than the inner diameter D′**1** of the multilayer hose **100**. In one embodiment, the outer diameter D**2** of the other end **2102** of the first connector **210** may be larger than the inner diameter D′**2** of the other multilayer hose **101**. Therefore, the multilayer hoses **100** and **101** may be joined tightly against the first connector **210**.
(40) According to one embodiment, as shown in FIGS. **4**A, **4**B, and **4**C, the sequence of processes by which connector assembly **200** is assembled may be as follows.
(41) In one embodiment, with reference to FIG. **4**A, the first connector **210** may be inserted into the inside of the second connector **220** through the opening **224** of the second connector **220**. Subsequently, in one embodiment not shown in the diagram, the multilayer hose **100** may be inserted into the end **2101** of the first connector **210** by being inserted into the inside of the second connector **220**. For example, one end **2101** of the first connector **210** may be inserted into the inside of the first rubber layer **110** of the multilayer hose **100**. In one embodiment, the first rubber layer **110** may be stably fixed to the first connector **210** as the outer diameter D**1** of the outermost end of the first connector **210** is formed larger than the inner diameter of the first rubber layer **110** D′**1**.
(42) Subsequently, though a swaging process illustrated in FIGS. **4**B and **4**C, one or more portions of the exterior of the second connector **220** may be pressurized in the direction of the first connector **210**. In one embodiment, with reference to FIG. **4**B, the second connector **220** may include a cover portion **221** covering the first connector **210**. While the multilayer hose **100** is inserted into the first connector **210**, the cover portion **221** of the second connector **220** of FIG. **4**A may be subjected to an external force F toward the multilayer hose **100** at multiple portions of the cover portion **221** as shown in FIG. **4**B. For example, the swaging mechanism used in the swaging process may compress a portion of the cover portion **221** of the second connector **220**.
(43) In one embodiment, with reference to FIG. **4**C, the second connector **220** may result in a repetition of concave and convex shapes through the swaging process performed per FIG. **4**B. In the following, the concave portion formed in the second connector **220** through the swaging process will be described as the pressurized portion **222**. The pressurized portion **222** may be a configuration formed by the swaging mechanism pressurizing the cover portion **221** toward the multilayer hose **100**. The convex shape of the second connector **220** is to be described as a protrusion **223**. The protrusion **223** may be part of the uncompressed cover portion **221** through the swaging mechanism in the swaging process.
(44) In one embodiment, the pressurized portion **222** of the second connector **220** may pressurize the exterior of the multilayer hose **100**. The multilayer hose **100** may be joined tightly to the first connector **210** by the exterior of the multilayer hose **100** being pressurized through the second connector **220** while the multilayer **100** is inserted into the first connector **210**.
(45) The following describes the detailed configuration of the first connector **210** and the second connector **220**.
(46) According to one embodiment, as shown in FIGS. **4**A to **4**C, and **5**A to **5**C, the first connector **210** may include a body **211**, a stopper **213** protruding from the body **211**, an inclined portion **214** located at the end **2101** of the body **211**, irregularities **215** consisting of the convex portion **2152** and the concave portion **2151**, and/or a contact portion **216** that is in contact with the serration **225** of the second stopper **213**. At least one of the above configurations may be omitted or at least one configuration may be added.
(47) In one embodiment, with reference to FIGS. **4**A to **4**C and **5**A to **5**C, the body **211** may be the main body of the first connector **210**. In one embodiment, the body **211** may include a second pipe **212**. In one embodiment, the first pipe **1001** of the multilayer hose **100** and the second pipe **212** of the body **211** may be connected as the body **211** is inserted into the inside of the multilayer hose

100.
(48) In one embodiment, with reference to FIGS. **4**A to **4**C and **5**A to **5**C, the stopper **213** may be formed by protruding along the outer periphery of the body **211**. In one embodiment, the stopper **213** may be a detent that prevents the second connector **220** from passing through the first connector **210** when the first connector **210** is inserted into the opening **224** of the second connector **220**, as shown in FIG. **4**A.
(49) In one embodiment, with reference to FIGS. **5**A to **5**C, the inclined portion **214** of the first connector **210** may be located at the end **2101** of the body **211**. In one embodiment, the end of body **211** may be the first portion to be inserted into the inside of the multilayer hose **100**. In one embodiment, the inclined portion **214** may be formed in a sloping form along the outer peripheral surface of the body **211** at the end of the body **211**. In one embodiment, the inclined portion **214** may consist of an uphill slope from the end of body **211** toward the stopper **213**. In one embodiment, the inclined portion **214** may be formed at various angles. For example, the angle θ**1** of the inclined portion **214** may be formed preferably about 11°-17° with respect to one side of the body **211**. One side of the body **211** may be flat. Therefore, the angle θ**1** of the inclined portion **214** may be formed about 11°-17° with respect to the plane.
(50) In one embodiment, with reference to FIGS. **4**A to **4**C and **5**A to **5**C, the outermost end of body **211** (e.g., the outermost end of the first connector **210**) may be the starting point of the inclined portion **214**. In one embodiment, the outer diameter D**1** of the outermost end of the body **211** may be about 15.9 mm to 16.1 mm. In addition, the end of the inclined portion **214** may be formed with an outer diameter greater than the outermost end of the body **211**. Here, the end of the inclined portion **214** may be the point at which the slope ends. For example, the outer diameter of the end of the inclined portion **214** may be about 16.8 mm to 17.2 mm. In one embodiment, the outer diameter of the outermost end of body **211** and the outer diameter of the end of the inclined portion **214** may be formed larger than the inner diameter D′**1** of the first rubber layer **110**. Therefore, the multilayer hose **100** may be joined tightly against the first connector **210**.
(51) In one embodiment, the slope of the included portion **214** may prevent oil flowing on the multilayer hose **100** from spilling between the first connector **210** and the multilayer hose **100**. For example, viscous oils may lose fluidity on sloping sections.
(52) In one embodiment, with reference to FIG. **4**C, the gap L between the first connector **210** and the pressurized portion **222** of the second connector **220** may be reduced as an inclined portion **214** is formed at the end of the first connector **210**. For example, if an inclined portion **214** is formed that constitutes an uphill slope from the end **2101** of the first connector **210** to the stopper **213**, the gap L between the end of the inclined portion **214** and the pressurized portion **222** of the second connector **220** may be reduced than the gap between the end of the inclined portion **214** and the pressurized portion **222** of the second connector **220** without an inclined portion **214** formed at the end of the first connector **210**. Therefore, compared to when the inclined portion **214** is not formed at the end of the first connector **210**, the multilayer hose **100** may be joined tightly against the first connector **210** by being compressed at the end of the first connector **210** through the end of the inclined portion **214** and the pressurized portion **222** of the second connector **220**. Therefore, the multilayer hose **100** may not be separated from the first connector **210** in an environment where vibrations are applied. In addition, oil leakage into the gap between the multilayer hose and the first connector **210** may be prevented.
(53) In one embodiment, with reference to FIGS. **4**A to **4**C and **5**A to **5**C, the first connector **210** may include irregularities **215** consisting of a convex portion **2152** and a concave portion **2151**. In one embodiment, irregularities **215** may form along the outer peripheral surface of the body **211**. In one embodiment, the concave portion **2151** may be a groove formed on one side along the outer periphery of the body **211** of the first connector **210**. The convex portion **2152** may be a part that is relatively protruded compared to the concave portion **2151**. In one embodiment, the irregularities **215** may prevent the body **211** from being damaged by dispersing the external force applied

through the pressurized portion **222** when one side of the body **211** is pressurized through the pressurized portion **222** of the second connector **220** in the swaging process.

(54) In one embodiment, with reference to FIGS. **5**A to **5**C, the width W**1** of the concave portion **2151** and the width W**2** of the convex portion **2152** may preferably be about 1.7 mm to 1.9 mm.

(55) In one embodiment, with reference to the second connector **220** of FIG. **4**C, the concave portion **2151** of the first connector **210** may be located correspondingly with the pressurized portion **222** of the second connector **220**. In one embodiment, in the swaging process, the swaging mechanism forming the pressurized portion **222** of the second connector **220** may be located corresponding to at least one of the plural concave portions **2151** of the first connector **210**. The pressurized portion **222** is formed in a position corresponding to the swaging mechanism, so that it may be located corresponding to at least one of the plural concave portions **2151** of the first connector **210**.

(56) In one embodiment, when the second connector **220** is compressed through the swaging mechanism while the multilayer hose **100** is inserted into the first connector **210**, if the swaging mechanism does not correspond to the concave portion **2151** of the first connector **210**, a portion of the multilayer hose **100** may be pressurized to the edge of the convex portion **2152** of the irregularities **215** and be damaged. According to one embodiment of this disclosure, when the multilayer hose **100** is pressurized through the swaging mechanism as the swaging mechanism is positioned to correspond to at least one of the plural concave portions **2151** of the first connector **210**, the multilayer hose **100** may be joined tightly against the concave portion **2151** of the irregularities **215**. In other words, as the swaging mechanism is positioned to correspond to at least one of the plural concave portions **2151** of the first connector **210**, the pressurized portion **222** of the second connector **220** formed through the swaging mechanism may press the multilayer hose tightly to the concave portion **2151** of the first connector **210**. Therefore, the multilayer hose **100** may be compressed to the first connector **210** without damage in the swaging process.

(57) According to one embodiment, as shown in FIGS. **4**A to **4**C and **6**A and **6**B, the second connector **220** may include a serration **225**. In one embodiment, the serration **225** may be a structure formed in a serrated or grooved shape on one side of the second connector **220** along the periphery of the opening **224** of the second connector **220**. In one embodiment, the serration **225** of the second connector **220** may be contacted with a contact portion **216** of the first connector **210**. In one embodiment, the contact portion **216** may be a groove formed along the outer periphery of the body **211** at a position adjacent to the stopper **213** of the first connector **210**. In one embodiment, the serration **225** may increase the cohesion between the configurations. For example, the second connector **220** may be joined tightly to the first connector **210** via the serration **225**. In one embodiment, the serration **225** may prevent the first connector **210** from rotating with respect to the second connector **220**.

(58) In one embodiment, the serration **225** may consist of a plurality of teeth **2251**. In one embodiment, the angle θ**2** between neighboring teeth **2251** may preferably be about 81.7°. In one embodiment, the height of the tooth **2251** may preferably be about 0.5 mm.

(59) According to one embodiment, the thickness of the multilayer hose **100** can be compressed to about 30%˜50% after being coupled to the connector assembly **200**. In one embodiment, the compression of the multilayer hose **100** may be calculated through mathematical equation 1.

(60) $\frac{b-a}{a} \times 100$ Equation1

a may be the thickness of the multilayer hose **100** before the first connector **210** is inserted into the inside the first rubber layer **110**. b may be the thickness of the multilayer hose in the state that the first connector **210** is inserted into the inside the first rubber layer **110** and the outer surface of the multilayer hose **100** is pressurized and compressed through the pressurized portion **222** of the second connector **220**.

(61) According to one embodiment, connector assemblies may be formed of a variety of materials. For example, the first connector **210** and the second connector **220** may be formed of a metal

material and/or a non-metallic material. Here, metal materials may include alloys such as aluminum, stainless steel (STS, SUS), iron, magnesium, titanium, etc., while non-metallic materials may include synthetic resins, ceramics, and engineering plastics.

(62) The connection structure of a hose for transmission oil of this disclosure may include a multilayer hose **100** and **101** including the first rubber layer **110** formed by the first pipe **1001**, the braided layer **120** formed on the outer surface of the first rubber layer and formed by braiding aramid fibers, and the second rubber layer **130** located on the outer surface of the first rubber layer; a first connector including a body inserted into the inside of the first rubber layer and including a second pipe connected with the first pipe, a stopper protruding from the body, and an inclined portion formed at the end of the body and forming an uphill slope of approximately 11°-17° to one side of the body toward the stopper; and a second connector **220** including a cover portion **221** covering the multilayer hose and a plurality of pressurized portions **222** formed concave with respect to the outer peripheral surface of the cover portion and in contact with the multilayer hose.

(63) In addition, the outer diameter D**1** of the outermost end of the body may be greater than the inner diameter D′**1** of the first rubber layer.

(64) In addition, the first rubber layer and the second rubber layer of the multilayer hose may be formed of any of the acrylic rubber (ACM, acrylate-chloroethyl vinylether monomer), ethylene acrylic rubber (AEM, ethylene acrylic elastomer), fluorinated rubber (FKM, fluorinated vinylidene monomer) and hydrogen nitrile rubber (H-NBR, hydrogenated acrylonitrile-butadiene rubber).

(65) In addition, the first connector may include irregularities **215** formed by repeating the convex portion **2152** and the concave portion **2151** on the outer peripheral surface of the body, and at least one of the plural pressurized portions of the second connector may be formed in a position corresponding to the concave portion of the first connector.

(66) In addition, the width of the concave portion W**1** and the width of the convex portion W**2** may be approximately 1.7 mm to 1.9 mm.

(67) In addition, the compression rate of the multilayer hose calculated according to the following mathematical equation 1 below may be 30%˜50%.

(68) $\frac{b-a}{a} \times 100$ Equation*l*

(a: the thickness of the multilayer hose before the first connector is inserted into the inside of the first rubber layer, b: the thickness of the multilayer hose in a state where the first connector is inserted into the inside of the first rubber layer and the outer surface of the multilayer hose is pressurized and compressed through the pressurized portion of the second connector).

(69) In addition, the braided layer may include a braid wound with a pitch of approximately 30 mm to 50 mm along the outer surface of the first rubber layer.

(70) In addition, the second connector may include a serration **225** including an opening **224**, and a tooth **2251** formed along the periphery of the opening and in contact with the first connector.

(71) In addition, the angle θ**2** between the neighboring teeth may be 81.7°.

(72) According to various embodiments disclosed in this document, it is possible to disclosure a multilayer hose that can be used under high pressure, extremely high temperature, and cryogenic condition and connect the transmission with the oil cooling device.

(73) According to various embodiments disclosed in this document, the connector assembly **200** connecting the multilayer hose **100** to another multilayer hose **101** may present a structure that adheres closely to the multilayer hose **100** and **101**. Therefore, the multilayer hose **100** and **101** and connector assembly **200** may maintain their coupling under external vibration. In addition, the adhesive structure of the multilayer hose **100** and **101** and the connector assembly **200** may prevent the oil flowing inside the multilayer hose **100** and **101** from spilling out to the outside of the multilayer hose **100** and **101**.

(74) It is understood that this disclosure considers and includes embodiments based on any two or more combinations of the embodiments disclosed and embodiments comprising any combination of the features of the disclosed embodiments, in addition to the embodiments disclosed above. That

is, the absence of an explicit indication that two features can be combined, or that two embodiments may be combined, does not mean that such a combination is not envisaged, but that such a combination must be seen as being included here.

EXPLANATION OF REFERENCE NUMERALS

(75) **100**: multilayer hose **110**: first rubber layer **120**: braided layer **130**: second rubber layer **210**: first connector **220**: second connector

Claims

1. A connection structure of a hose for transmission oil comprising: a multilayer hose comprising: a first rubber layer formed by a first pipe; a braided layer formed on an outer surface of the first rubber layer and formed by braiding aramid fibers; and a second rubber layer located on the outer surface of the first rubber layer; a first connector comprising: a body inserted into an inside of the first rubber layer; a second pipe connected with the first pipe; a stopper protruding from the body; and an inclined portion formed at an end of the body and forming an uphill slope of approximately 11° to 17° to one side of the body toward the stopper; and a second connector comprising: a cover portion covering the multilayer hose; and a plurality of pressurized portions formed concave with respect to an outer peripheral surface of the cover portion and in contact with the multilayer hose; wherein the first connector comprises irregularities formed by repeating convex portions and concave portions on an outer peripheral surface of the body; and at least one of the plurality of pressurized portions of the second connector is formed in a position corresponding to one of the concave portions of the first connector; and wherein an end of the inclined portion protrudes in a radial direction more than the convex portions from the outer peripheral surface of the body.
2. The connection structure of a hose for transmission oil of claim 1, wherein an outer diameter of an outermost end of the body is greater than an inner diameter of the first rubber layer in a state prior to application of elastic force.
3. The connection structure of a hose for transmission oil of claim 1, wherein the first rubber layer and the second rubber layer of the multilayer hose are formed of one of the following materials: acrylate-chloroethyl vinyl ether monomer, ethylene acrylic rubber, fluorinated vinylidene monomer, and hydrogenated nitrile rubber.
4. The connection structure of a hose for transmission oil of claim 1, wherein a width of the concave portion and the convex portion of the first connector is approximately 1.7 mm to 1.9 mm, respectively.
5. The connection structure of a hose for transmission oil of claim 1, wherein a compression ratio of the multilayer hose calculated according to mathematical Equation 1 below is approximately 30% to 50% $\frac{b-a}{a} \times 100, \quad \text{Equation} \, l$ wherein: a: a thickness of the multilayer hose before the first connector is inserted into the inside of the first rubber layer; and b: a thickness of the multilayer hose in a state where the first connector is inserted into the inside of the first rubber layer and an outer surface of the multilayer hose is pressurized and compressed through one of the plurality of pressurized portions of the second connector.
6. The connection structure of a hose for transmission oil of claim 1, wherein the braided layer comprises a braid wound with a pitch of approximately 30 mm to 50 mm along the outer surface of the first rubber layer.
7. The connection structure of a hose for transmission oil of claim 1, wherein the second connector comprises a serration comprising an opening and a tooth formed along an outer periphery of the opening and in contact with the first connector.
8. The connection structure of a hose for transmission oil of claim 7, wherein the serration

comprises a second tooth, wherein the opening is between and adjacent to the tooth and the second tooth, wherein an angle between the tooth and the second tooth is 81.7°.